Assignment 1: Variables, Data Types, and Console Operations

Task: Declare variables using var, let, and const to store a user's name, age, employment status, and a list of skills. Use typeof to check each variable's type and log them to the console.

JavaScript Code

```
var name = "Alice";
let age = 25;
const isEmployed = true;
const skills = ["HTML", "CSS", "JavaScript"];

console.log(`Name: ${name} (Type: ${typeof name})`);
console.log(`Age: ${age} (Type: ${typeof age})`);
console.log(`Is Employed: ${isEmployed} (Type: ${typeof isEmployed})`);
console.log(`Skills: ${JSON.stringify(skills)} (Type: ${typeof skills})`);
```

Expected Output

```
Name: Alice (Type: string)
Age: 25 (Type: number)
Is Employed: true (Type: boolean)
Skills: ["HTML","CSS","JavaScript"] (Type: object)
```

Assignment 2: Operators, Expressions, and Input Handling

Task: Use prompt() to capture a user's bill amount and tip percentage. Perform explicit type conversion (e.g., using Number()), compute the total using arithmetic and assignment operators, and log a formatted string.

JavaScript Code

```
const billInput = prompt("Enter bill amount:", "100");
const tipInput = prompt("Enter tip percentage:", "15");

const subtotal = Number(billInput);
const tipPercentage = Number(tipInput);
const tip = subtotal * tipPercentage / 100;
let total = subtotal;
total += tip;

console.log(`Subtotal: $${subtotal}`);
console.log(`Tip Percentage: ${tipPercentage}%`);
console.log(`Calculated Tip: $${tip}`);
console.log(`Total Amount to Pay: $${total}`);
```

Expected Output

```
Subtotal: $100
Tip Percentage: 15%
Calculated Tip: $15
Total Amount to Pay: $115
```

Assignment 3: Conditional Logic Implementation

Task: Create a script that evaluates a user's numerical test score using if...else statements, determines a letter grade using a switch statement, and uses a ternary operator to print whether they passed or failed.

JavaScript Code

```
const score = 85;
let grade;

if (score >= 90) {
  grade = "A";
} else if (score >= 80) {
  grade = "B";
} else if (score >= 70) {
  grade = "C";
} else if (score >= 60) {
  grade = "D";
} else {
  grade = "F";
}

switch (grade) {
  case "A": console.log("Excellent"); break;
  case "B": console.log("Good"); break;
  case "C": console.log("Average"); break;
  case "D": console.log("Needs Improvement"); break;
  default: console.log("Failing grade");
}

const status = score >= 40 ? "Passed" : "Failed";
console.log(`Score: ${score}`);
console.log(`Grade: ${grade}`);
console.log(`Status: ${status}`);
```

Expected Output

```
Good
Score: 85
Grade: B
Status: Passed
```

Assignment 4: Loop Control Structures

Task: Write a for loop to print a multiplication table up to 5. Use a while loop with a break statement to find the first number greater than 10 that is divisible by 6, and a do...while loop that skips numbers using continue.

JavaScript Code

```
console.log("-- Multiplication Table of 5 --");
for (let i = 1; i <= 5; i++) {
  console.log(`5 x ${i} = ${5 * i}`);
}

console.log("\\n-- First number > 10 divisible by 6 --");
let number = 11;
while (true) {
  if (number > 10 && number % 6 === 0) {
    console.log(`Found: ${number}`);
    break;
  }
  number++;
}

console.log("\\n-- do...while with continue (skip 3) --");
let count = 0;
do {
  count++;
  if (count === 3) continue;
  console.log(count);
} while (count < 5);
```

Expected Output

```
-- Multiplication Table of 5 --
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25

-- First number > 10 divisible by 6 --
Found: 12

-- do...while with continue (skip 3) --
1
2
4
5
```

Assignment 5: Function Architecture and Scope

Task: Define a standard function and an arrow function that accept parameters (e.g., length and width) with default values to calculate the area of a rectangle. Demonstrate function and block scope limits.

JavaScript Code

```
function rectangleArea(length = 5, width = 4) {
  return length * width;
}

const rectangleAreaArrow = (length = 5, width = 4) => length * width;

console.log(`Standard Function Area (Default 5x4): ${rectangleArea()}`);
console.log(`Arrow Function Area (Custom 10x3): ${rectangleAreaArrow(10, 3)}`);

function testScope() {
  var functionVar = "available inside function";
  if (true) {
    let blockVar = "available inside this block";
    const blockConst = "also block-scoped";
    console.log(blockVar);
    console.log(blockConst);
  }
  console.log(functionVar);
  // console.log(blockVar); // ReferenceError: blockVar is not defined
}

testScope();
```

Expected Output

```
Standard Function Area (Default 5x4): 20
Arrow Function Area (Custom 10x3): 30
available inside this block
also block-scoped
available inside function
// Accessing blockVar outside the if block causes:
// ReferenceError: blockVar is not defined
```

Assignment 6: Arrays, Objects, and Core Methods

Task: Create an array of product objects (with name and price properties). Use forEach() to print each product, filter() to find items under a certain price, and map() to create an array of formatted name-price strings.

JavaScript Code

```
const products = [
  { name: "Laptop", price: 800 },
  { name: "Mouse", price: 25 },
  { name: "Keyboard", price: 45 }
];

console.log("-- All Products (forEach) --");
products.forEach(product => {
  console.log(`Item: ${product.name} - Price: $${product.price}`);
});

console.log("\\n-- Filtered Products (Price < $50) --");
const affordableProducts = products.filter(product => product.price < 50);
console.log(affordableProducts);

console.log("\\n-- Mapped Product Tags (map) --");
const productTags = products.map(
  product => `${product.name.toUpperCase()} costs $${product.price}`
);
console.log(productTags);
```

Expected Output

```
-- All Products (forEach) --
Item: Laptop - Price: $800
Item: Mouse - Price: $25
Item: Keyboard - Price: $45

-- Filtered Products (Price < $50) --
[ { name: 'Mouse', price: 25 }, { name: 'Keyboard', price: 45 } ]

-- Mapped Product Tags (map) --
[ 'LAPTOP costs $800', 'MOUSE costs $25', 'KEYBOARD costs $45' ]
```